

Minseo Kwon

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RESEARCH INTERESTS

Robotics, Task and Motion Planning (TAMP), Robot Learning

EDUCATION

Princeton University, Princeton, NJ

Ph.D. in Electrical and Computer Engineering

Jul 2026 – May 2031 (Expected)

- Advisor: Dr. Tom Silver

Ewha Womans University, Seoul, Korea

M.S. in Computer Science and Engineering

Sept 2024 – Aug 2026

- Advisor: Dr. Young J. Kim
- Thesis: *Neuro-Symbolic Task and Motion Planning with Kinodynamic Constraints using Vision Language Models*

B.S. in Computer Science and Engineering & Mathematics

Mar 2020 – Aug 2024

- Graduated with Honors: Magna Cum Laude

PUBLICATIONS

Conference Papers

[C03] M. Kwon and Y. J. Kim, **Kinodynamic Task and Motion Planning using VLM-guided and Interleaved Sampling**, *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [🔗](#) [🌐](#) [📄](#)

[C02] M. Kwon, Y. Kim, and Y. J. Kim, **Fast and Accurate Task Planning using Neuro-Symbolic Language Models and Multi-level Goal Decomposition**, *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [🔗](#) [🌐](#) [📄](#)

[C01] S. Kwon*, M. Kwon*, H. Kim* and J. Sim, **ToMato: Accelerating ViT via Token Merging**, *Autumn Annual Conference of Institute of Electronics and Information Engineers (IEIE)*, 2023. (* Equal Contribution) [🔗](#) [🌐](#) [📄](#)

Journal Papers

[J02] V. De Gusseme, T. Lips, R. Proesmans, J. Hietala, G. Lee, J. Choi, J. Choi, G. Kim, P. Yonrith, D. Tabernik, A. Gams, P. Nimac, M. Urbas, J. Muhovič, D. Skočaj, M. Mavsar, H. Yu, M. Kwon, et al., **A Dataset and Benchmark for Robotic Cloth Unfolding Grasp Selection: The ICRA 2024 Cloth Competition**, *The International Journal of Robotics Research*, 2025. [🔗](#)

[J01] M. Kwon, and Y. J. Kim, **Neuro-Symbolic Task Replanning using Large Language Models**, *The Journal of Korea Robotics Society*, 2025. [🔗](#)

Workshop Papers

[W01] M. Kwon, and Y. J. Kim, **Neuro-Symbolic Task Planning and Replanning using Large Language Models**, *ICRA 2025 Workshop on Foundation Models and Neuro-Symbolic AI for Robotics*, 2025. [🔗](#)

RESEARCH EXPERIENCE

Ewha Womans University, Seoul, Korea

Graduate Research Assistant (Advisor: Dr. Young J. Kim)

Sept 2024 - Jun 2026

- Developed a kinodynamic TAMP framework that leverages VLMs as forward and backtracking heuristics, improving success rates and planning time compared to traditional and LLM-based TAMP planners. [C03]

Undergraduate Research Assistant (Advisor: [Dr. Young. J. Kim](#)) *Dec 2022 – Aug 2024*

- Developed a neuro-symbolic task planner that leverages LLMs to decompose large-scale problems into smaller sub-problems, improving planning efficiency and accuracy. [\[C02\]](#) [\[J01\]](#) [\[W01\]](#)
- Developed a robotic cloth unfolding pipeline using a point cloud-based sharp edge detection method for grasp localization, winning 3rd place at the [ICRA 2024 Cloth Competition](#). [\[H03\]](#) [\[J02\]](#) [🔗](#)

Capstone Design Project (Advisor: [Dr. Jaehyeong Sim](#)) *Jan 2023 – Dec 2023*

- Developed a model compression framework by recursively merging tokens in early Transformer blocks, reducing ViT inference latency by 22.19% while preserving over 80% accuracy. [\[H01\]](#), [\[H02\]](#) [\[C01\]](#)

HONORS & AWARDS

[H04]	Finalist , What Bimanuals Can Do Challenge (WBCD), ICRA	<i>2025</i>
[H03]	3rd Place , ICRA 2024 Cloth Competition	<i>2024</i>
[H02]	Silver Prize , Ewha Engineering Capstone Design Contest	<i>2023</i>
[H01]	3rd Place , Undergraduate Research Paper Contest, Autumn Annual Conference of IEIE	<i>2023</i>

SCHOLARSHIPS & ACADEMIC AWARDS

[S04]	Research Assistant Scholarship , Ewha Womans Univ.	<i>2025</i>
[S03]	Graduate Admissions Scholarships , Ewha Womans Univ.	<i>2024 - 2025</i>
[S02]	Dean's List (6 semesters), Ewha Womans Univ.	<i>Fall 2020 - Fall 2023</i>
[S01]	Undergraduate Admissions Scholarship , Ewha Womans Univ.	<i>2023</i>
• Ranked 1st among ~2,970 applicants in the Science and Engineering entrance exam.		

SKILLS

Programming Languages	C++, C, Java, Python, MATLAB
Robot SW	ROS, Genesis, CoppeliaSim, Mujoco, Gazebo, PyBullet
Robot HW	UR5e, Robotiq 3F Adaptive Gripper
Languages	Korean, English

TEACHING

Altu-Bitu, Algorithm Tutoring Program, Ewha Womans Univ.	<i>Fall 2022</i>
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REVIEWING

IEEE International Conference on Robotics and Automation (ICRA)	<i>2026</i>
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